College of Computing
Computer Science and Artificial Intelligence

Darp : AI-Based Early Risk Detection in Crowded and High-Temperature Environments

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ABSTRACT

Large-scale mass gatherings such as the Hajj season present significant health and safety challenges due to extreme crowd density and elevated temperatures, increasing the risks of heat exhaustion and fainting. Current solutions often address crowd management and health monitoring separately, creating a gap in integrated real-time risk prevention.

This project proposes a lightweight AI-based prototype for early risk detection in crowded and high-temperature environments. The system employs a Convolutional Neural Network (CNN), trained on the ShanghaiTech dataset, to classify crowd density with a target accuracy of at least 85%. Crowd density is integrated with normalized temperature values and user-reported symptoms through a rule-based engine to compute a weighted baseline risk score.

The system generates location-based alerts to support medical teams in prioritizing rapid intervention in high-risk zones. It also provides preventive notifications for pilgrims within the same area without offering medical diagnosis or storing personal data.

The expected outcome is a fully functional web-based prototype that enables spatial risk monitoring and supports timely decision-making while preserving privacy.

Keywords: Hajj Safety, Crowd Density Estimation, Heat Risk Assessment, Early Warning System.

1. BACKGROUND AND CONTEXT

This project falls within the field of Computer Science, specifically Artificial Intelligence (AI), focusing on Computer Vision based on Deep Learning methods for crowd and health safety applications. The project aims to use image analysis to estimate crowd density and then combine this information with heat conditions and simple health-related symptoms to enable early detection of potential risks. In large-scale mass gatherings such as Hajj, high crowd density and elevated temperatures occur simultaneously,significantly increasing the risk of heat exhaustion, fainting, and other heat-related health problems [5, 6]. Early warning signs, such as dizziness or fatigue, are often difficult to detect in crowded and fast-moving environments, which may delay timely response and increase overall risk. Previous studies indicate that existing solutions address these challenges in isolation. Crowd analysis systems focus on estimating density and movement patterns to support crowd management and safety planning [3, 4, 7, 8], but they do not link this information to health-related risks. In contrast, digital health systems, such as symptom-checker applications, rely on user-reported symptoms and emphasize user experience, often neglecting environmental factors such as heat exposure and crowd density [1, 2]. Overall, the current state of research remains fragmented, with crowd analysis and health monitoring systems developed as separate solutions rather than within a unified, context-aware framework. This reveals a clear research gap in the lack of an integrated system that combines crowd density analysis, heat conditions, and symptom inputs to support early risk warning. This project addresses this gap by proposing a lightweight AI-based system that supports collective decision-making without identifying individuals or storing sensitive medical data, making it suitable for large-scale mass-gathering environments such as Hajj.

2. PROJECT SCOPE

This project aims to develop an AI-based prototype to support early health and safety monitoring in crowded and high-temperature environments. The system focuses on analyzing heat- and crowd-related risk indicators using simple user-reported symptoms and general visual cues extracted from images, such as visible fatigue or skin redness. The project does not provide medical diagnosis, does not store personal data, and does not address chronic or complex medical conditions. It is limited to simple and potentially urgent cases related to heat and crowd exposure and will be implemented as a prototype.

3. PROJECT DESCRIPTION AND IDENTIFIED GAP

The Hajj environment suffers from many complex health and security challenges due to crowd density and high temperatures. Previous research has extensively explored crowd density using computer vision and deep learning [3, 4, 7], while other studies have focused on heat-related illnesses through clinical measurements and post-accident diagnosis [5, 6], symptom-based health systems have been studied mainly from a user experience perspective without incorporating environmental or crowd density-related risk factors [1, 2]. It was often developed independently, resulting in solutions that lack integration. Studies highlight a critical gap represented by the absence of integrated preventive systems that combine environmental risks associated with heat, congestion conditions and early health indicators in one framework suitable for high-congested environments such as Hajj.Current solutions focus on diagnostic rather than early warning and require complex clinical data or wearable devices and lack immediate spatial awareness and decision support mechanisms for medical response teams. The project aims to fill these gaps by developing an intelligent platform to monitor the health and safety of pilgrims in crowded and high-temperature pilgrimage environments. The proposed solution combines crowd density analysis using computer vision technology with a hypothetical analysis of heat risks and simple symptoms reported by pilgrims themselves, such as dizziness, fatigue or itching. The proposed system follows a hierarchical integration approach. First, a CNN-based crowd density model, drawing on recent studies [3, 7, 8], classifies each zone's density as low, medium, or high. Second, a heat risk simulation module computes a baseline environmental risk score using the formula: $(\text{Density} \times 0.6) + (T_{norm} \times 0.4)$, where density is assigned a higher weight (60%) due to its dual impact on both mobility restriction and heat stress. Third, a rule-based engine incorporates self-reported symptoms by adding +1 to the score for a single report, and +2 for multiple reports (three or more) in the same geographical zone. The final score

determines the alert level: low (1.0-1.5), preventive warning (1.6-2.0), high risk (2.1-2.6), or critical (>2.6), which triggers an immediate notification to medical teams. The controlled environment is divided into pre-defined areas, allowing the system to link reported cases to specific locations (e.g. Zone A, Zone B). Spatial division helps identify high-risk areas, and supports medical teams by providing location-based alerts and aggregated risk information. The system does not provide a medical diagnosis but focuses on early warning and decision support to help reduce health risks, prioritize medical response and enhance public safety during the Hajj season .

4. PROJECT OBJECTIVES

The project aims to:

1. Design and develop an intelligent software platform capable of monitoring health risks resulting from crowd congestion and high temperatures during the Hajj season.
2.Employ computer vision techniques to monitor crowd density and identify high- risk zones, targeting a classification accuracy of at least 85% when tested on benchmark datasets, consistent with findings reported in recent literature [3, 4].
3. Provide a hypothetical representation of heat stress risks based on crowd density levels and environmental conditions, using a formula that weights density more heavily (60%) due to its dual impact on both heat stress and mobility restriction, and 40% for normalized temperature.
4. Link crowd density and heat risks to self-reported symptoms through a rule-based engine that dynamically updates zone risk scores, triggering immediate notifications to medical teams when critical levels are detected.
5. Offer a simple user interface that accommodates diverse pilgrim demographics and supports medical teams in decision-making without requiring advanced medical diagnosis, with an end-to-end response time target (from image analysis to alert generation) of under 5 seconds in the prototype system.
6. Enable spatial risk monitoring by dividing Hajj ritual areas into predefined sectors, allowing medical teams to prioritize responses and allocate resources more efficiently.
7.Evaluate the integrated system's performance in a simulated environment by measuring the accuracy of risk level classification compared to validated reference scenarios, precision and recall rates in detecting high-risk areas, and user satisfaction (medical teams) regarding alert clarity through a simplified questionnaire.

5. PROJECT TIMELINE

Week	Activity
1	Defining project scope and write project description
2	Review relevant background and define project goals
3	Literature search: collecting relevant articles and papers
4	Literature review: read, summarize findings, and identify gaps
5	Proposal finalizing and submission

6. EXPECTED OUTCOMES

By the end of our project, several tangible and measurable deliverables will be produced. First, a fully functional web-based AI prototype will be developed to support early health and safety monitoring of crowded and high- temperature Hajj holy sites such as Mina, Arafat, and Muzdalifah. The web-based AI prototype will include a user interface for pilgrims to report their simple symptoms and an administrative interface to visualize the crowd density level, simulated heat level, and zone-based alerts. Second, the project will provide a trained and evaluated Convolutional Neural Network (CNN) model for crowd density classification (low, medium, and high). The model will be trained on a suitable dataset and then tested using performance metrics. The classification accuracy of the model will be at least 85%. The model will also be fully integrated with the web-based system to perform real-time density estimation from video frames. Third, a heat risk simulation model will be implemented based on the weighted formula (Density × 0.6) + (T_{norm}× 0.4) The model will provide categorized alert levels, including low, preventive warning, high risk, and critical. Furthermore, a rule-based symptom integration engine will be developed to dynamically update the risk score based on the number of user-reported symptoms within the same zone. The system will also implement spatial segmentation of Hajj areas into predefined zones to enable location-based monitoring and prioritization of medical response. The prototype will aim to achieve an end-to-end response time of under five seconds, from image analysis to alert generation. The system will generate administrative alerts for medical teams to support rapid intervention in high- risk zones. In addition, it will provide zone-based preventive notifications for pilgrims located in high-risk areas. These alerts are informational and non-diagnostic, encouraging reduced physical effort and increasing awareness of environmental risks.

7. PROPOSED METHODOLOGY OR APPROACH

This project follows an applied development methodology to build a web-based AI prototype for early health risk monitoring in crowded and high-temperature environments such as Hajj. The system integrates computer vision– based crowd density classification, heat-risk representation, and simple self-reported symptoms within predefined spatial zones. The platform is implemented using Python, with OpenCV for image processing and CNN models developed using TensorFlow or PyTorch to classify density levels (low, medium, high). The crowd density model is trained using the publicly available ShanghaiTech Part A dataset [11], which contains high-density crowd scenes representative of large-scale gatherings such as Hajj. The output of the CNN-based model is mapped into numerical density levels (Low = 1, Medium = 2, High = 3), which are then used in the risk calculation formula. A lightweight web interface is developed using Flask or Streamlit. The crowd density model is evaluated using accuracy, precision, recall, and F1-score. The dataset is split into 70% training, 15% validation, and 15% testing to ensure proper generalization. Environmental temperature values are obtained from historical weather data for Makkah or simulated within realistic Hajj temperature ranges (30–45°C).Heat-related risk is calculated using a weighted formula: Baseline Risk Score = 0.6 (Density) + 0.4 (T_{norm}) This model follows a weighted composite indicator approach as described in the OECD Handbook on Constructing Composite Indicators [10], where multiple factors are linearly combined using predefined weights. Crowd density is assigned a higher weight (0.6) due to its amplified impact on mobility restriction and heat stress under mass-gathering conditions, while temperature contributes 0.4 as an environmental stress factor supported by heat-related health studies [9]. where temperature values are normalized using: T_{norm} = (T − T_min) / (T_max − T_min) This ensures temperature values are scaled between 0 and 1 before integration into the risk model. To incorporate health-related indicators, a rule-based symptom engine adjusts the risk score by adding +1 for a single symptom report and +2 for multiple reports (three or more) within the same geographical zone. Self-reported symptoms (e.g., dizziness, fatigue, itching) are collected through the system interface and used only to adjust the risk score without storing personal information. The final score is categorized into predefined alert levels (low, medium, high, or critical), which are displayed through an administrative monitoring interface to support medical teams in prioritizing response efforts in high-risk zones .

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